

### Unit 1 Practice Questions

1. Write and illustrate electron configuration of all elements from H ( $Z=1$ ) to Zn ( $Z=30$ ).
2. Write a brief note about Mendeleev's and Moseley's periodic table.
3. Predict the period, block and group of P ( $Z=15$ ), Fe ( $Z=26$ ), Rb ( $Z=37$ ), Ag ( $Z=47$ ).
4. Calculate  $Z_{\text{eff}}$  on outermost s electron for K ( $Z=19$ ) and Co ( $Z=27$ ).
5. Calculate  $Z_{\text{eff}}$  on 3d electron for V ( $Z=23$ ) & Ni ( $Z=28$ ).
6. Definitions- Atomic radius, effective nuclear charge, ionization energy, electron affinity and electronegativity.
7. Rank  $\text{P}^{3-}$ ,  $\text{Ca}^{2+}$ ,  $\text{Cl}^-$ , Ar and  $\text{K}^+$  in ascending order of radius along with justification.  
Atomic nos: P=15, Ca=20, Cl=17, Ar=18, K=19
8. Arrange Si ( $Z=14$ ), P ( $Z=15$ ), S ( $Z=16$ ) and Na ( $Z=11$ ) in descending order of ionization energy along with reason.
9. Why is nitrogen's electron affinity less than oxygen. Give proper reason.
10. Rank K ( $Z=19$ ), Si ( $Z=14$ ), O ( $Z=8$ ), F ( $Z=9$ ) in ascending order of electronegativity with proper justification.